Product backlog

Product objectives:

Offer management a simple-to-use web-based dashboard that will allow for monitoring the static fleet of electric vehicles and the estimation of the cost of charging during a certain period.

Priority:

Prioritize tasks based on delivery priority. US-01 and US-02 are the main functionalities, while all others are used to improve the quality of work and make it ready for submission.

ID	User story	Acceptance criteria
US-01	As a fleet manager, I need to see the fleet and their battery levels and status so that I can know which vehicles are available for work.	Show 5 to 10 EVs that are pre-defined; the existing prototype currently has 10 EVs. The rows have information such as Vehicle ID, Vehicle Model, Battery(%) and Status(Available/Charging/On Trip).
US-02	Being a fleet manager, my need is to choose the vehicle and also specify the charging hours to calculate the cost of charging.	The list should be populated with the vehicles in the fleet. The constant calculation is hours x 7.5 kW x rate for each kilowatt-hour. Show the selected car and cost estimate in USD.
US-03	Invalid charging times should be excluded as a fleet manager because I don't need a false estimation.	Reject all invalid data with a specific error message. Accept valid positive numbers as hours. Reset an existing estimate whenever its associated vehicle or hours have changed.
US-04	As a member of the team, I require calculator tests so as to ensure that the important calculations work correctly.	Test the cost function using unit tests with positive values for hours; reject negative and non-numeric values. The test results are satisfactory before submission.
US-05	As a reviewer, I need to be provided with proper instructions to run the prototype to learn about the tests.	README tells you how to launch the app and test its calculator. Describe one possible bug and how it was prevented by coding/testing.
US-06	As a reviewer, I need a small demo in order to check the functionality and collaboration aspects.	Five-minute-long video presents information about the fleet overview, charger calculator, repository branches, meaningful commits, and real pull requests.

Sprint deliverables support

Allocation of sprint tasks; 150-word sprint retrospective; 200-word security memorandum on two security vulnerabilities, effects and countermeasures; and 200-word ACM Ethics memorandum on privacy or bias issues.

Basis: Provided ICS432 Assignment 2, pp. 1-5 and provided EcoCharge prototype. GitHub and pull requests adhere to agreed-upon repository process by the team.